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Thermodynamic properties of fission products (Pr, Ce, Er) in liquid bismuth by thermodynamic assessment

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HIGHLIGHTS

- Bi-Pr, Bi-Ce, and Bi-Er systems were modelled utilizing the CALPHAD technique.
- Experimental data in Bi-Pr, Bi-Ce, and Bi-Er systems were well reproduced.
- Thermodynamic properties of intermetallic compounds and liquid were calculated.
- The activity coefficients of fission products in liquid bismuth were calculated and analyzed.

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ABSTRACT

The binary systems of Bi-Pr, Bi-Ce, and Bi-Er were thermodynamically assessed by the CALPHAD method based on the available experimental data utilizing sub-regular model and the two-sublattice model. Enthalpies and entropies of formation of intermetallic compounds and enthalpies of mixing of liquid phase at different concentrations were calculated. The activity coefficients of fission products in liquid bismuth at different concentrations and temperatures were also obtained, which supplements the existing data that are limited to dilution cases. Additionally, the change patterns of the activity with concentration of these three systems were analyzed and discussed. Based on the phase diagram assessed in the study, the correlations for the temperature-dependent solubility of the fission products in liquid bismuth were also developed. The present study provides fundamental data in a broad concentration range when liquid bismuth is applied as extraction media for the separation of fission products from a molten salt.

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1. Introduction

Due to its low melting point and immiscibility with molten salts, liquid bismuth has been widely used in the nuclear field for the separation of the actinides and fission products from molten salts in the applications of pyroprocessing for used fuel treatment and nuclear coolant cleanup for molten salt reactors. Liquid bismuth can be used as the cathode in pyroprocessing. In this case, the spent fuels at the anode are dissolved into the molten LiCl-KCl salt, and the soluble cations of uranium, plutonium, and the minor actinides [1] as well as the fission products [2] can be electro-deposited into the liquid bismuth cathode by the electrolysis:

$$Anode: M \rightarrow M^{n+}(in\ salt) + ne^- \quad (1)$$

$$Cathode: M^{n+}(in\ salt) + ne^- \rightarrow M(in\ Bi) \quad (2)$$

The separation of the actinides and fission products can also be fulfilled by the multistage reductive extraction process [3]. The actinides and fission products dissolved in the molten LiCl-KCl salt are reduced successively by contacting the lithium activated liquid bismuth according to [4]:

$$M^{n+}\ (in\ salt) + nLi(in\ Bi) \leftrightarrow M(in\ Bi) + nLi^+(in\ salt) \quad (3)$$

In many molten salt fueled nuclear reactor concepts, the

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reprocessing of the fuel salt involves the selective reduction of actinides and rare earth fission products from molten fluorides to liquid bismuth according to reaction (3). Liquid bismuth with dissolved lithium or thorium has been selected as the extraction media for the Molten Salt Breeder Reactor [5], Molten Salt Actinide Recycler and Transforming system [6], Non-moderated Thorium Molten Salt Reactor [7], and Molten Salt Fast Reactor (MSFR) [8]. The extracted actinides in the bismuth pool can be recovered back to the cleaned molten fluorides by electrolysis and sent back to the reactor core, while the lanthanides in the bismuth are back-extracted to the molten chloride medium by electrolysis for disposal. The bismuth based separation process is also being considered for the cleanup of the primary coolant salt of the molten salt cooled reactor concepts such as Fluoride Salt-Coolant High-Temperature Reactor [9,10]. The leached radionuclides in the coolant salt are removed by reductive extraction using lithium as reducing agent, or directly by electrolysis.

All the applications of liquid bismuth require the thermodynamic data of actinides and fission products in bismuth. For example, the activity coefficient of actinides and fission products in the bismuth is an important factor for both the chemical extraction and the electrolysis process. For the chemical extraction (reaction (3)), the distribution of M (e.g., actinides or fission products) in the salt and liquid bismuth is dependent on the activity coefficient of M in bismuth according to equation (4). For the selective reduction using electrolysis, the potential is a key factor which is also related to the activity coefficient of M in Bi, as displayed in equation (5):

$$K = \frac{a^n_{Li^+} a_M}{a^n_{Li} a_{M^{n+}}} = \frac{X^n_{Li^+} X_M}{X^n_{Li} X_{M^{n+}}} \frac{\gamma^n_{Li^+} \gamma_M}{\gamma^n_{Li} \gamma_{M^{n+}}} \tag{4}$$

$$E_{eq} = E^{\circ}_{M^{n+}/M} + \frac{RT}{nF} \ln\left(\frac{X_{M^{n+}} \gamma_{M^{n+}}}{X_M \gamma_M}\right) \tag{5}$$

where K is the equilibrium constant; E_{eq} and E° are equilibrium potential and standard potential, respectively; a, X and γ denote the activity, the mole fraction and the activity coefficient, respectively.

Consequently, massive studies have been conducted to study the activity coefficients of fission products in liquid bismuth [11–15]. However, most of them (if not all) studied the activity coefficients in the infinite dilutions or were based on the assumption of the infinite dilutions. Recently, there are some studies investigating the concentration dependence of fundamental data of elements in LiCl-KCl molten salt [16–19], which have shown the strong concentration dependence of the activity coefficient. For the practical applications of bismuth, the infinite dilution assumption is not applicable. Therefore, it is necessary to study the activity coefficients and other fundamental properties of fission products in liquid bismuth at different concentrations rather than limiting to the dilution situation.

CALPHAD (CALculation of PHAse Diagrams) is a powerful method to analyze thermodynamic properties of the binary systems with fission products and bismuth. Several bismuth-fission products systems have been thermodynamically studied by the method of the CALPHAD such as Bi-La [20], Bi-Nd (Tm) [21], Bi-Tb [22], Bi-Gd (Dy, Ho, Er) [23,24], Bi-Tm [25] and Bi-Yb [26]. Pr, Ce, and Er are three important fission products. Thermodynamic assessment of Bi-Pr which has never been reported before was conducted in the present study. Li et al. [27] and Wang et al. [28] did the thermodynamic assessments of Bi-Ce and Bi-Er, but the optimizations of the solid solutions were not considered. Thus, the Bi-Ce and Bi-Er systems were also studied in the present work with an improved optimization. In addition to the phase diagram optimization as done in available studies, the following fundamental data was obtained in the present study: enthalpy of formation, entropy of formation, enthalpy of mixing, concentration-dependent activity coefficient and activity, and temperature-dependent solubility for all the three systems.

2. Thermodynamic models

The liquid and solid solution (e.g. BCC, FCC, HCP) were described using a sub-regular solution model based on the random mixing of the constituent atoms with the Redlich-Kister formula [29]. The Gibbs energy of liquid phase can be expressed by

$$G^{liq} = \sum_{i=A,B} x_i G_i^{liq,0} + RT \sum_{i=A,B} x_i ln x_i + x_A x_B \sum_{n=0}^{m} L_n^{liq} (x_A - x_B)^n \tag{6}$$

where $G_i^{liq,0}$ is the Gibbs free energy of pure component i at liquid state taken from SGTE database [30], x_i is the mole fraction of component i, R is the gas constant, T is the temperature in Kelvin. L_n^{liq} is the binary interaction parameter and can be expressed by

$$L_n^{liq} = a_n + b_n T + c_n T ln T \tag{7}$$

where the coefficients a_n, b_n and c_n are the parameters that would be optimized in the present work based on experimental data using Thermo-Calc software [31].

The intermetallic compounds in the Bi-Pr, Bi-Ce, and Bi-Er binary systems were treated as stoichiometric compounds because of their narrow homogeneity [27,28]. The Gibbs free energy of these intermetallic compounds can be written by the two-sublattice model [32], it was formulated as

$$G^{A_m B_n} = \frac{m}{m+n} G_A^{SER,0} + \frac{n}{m+n} G_B^{SER,0} + a + bT \tag{8}$$

where m and n are the stoichiometric coefficients of the compound $A_m B_n$, $G_A^{SER,0}$ and $G_B^{SER,0}$ are the molar Gibbs free energies of the pure components A and B at their standard element reference states, respectively. a and b are the parameters to be optimized in the present work.

3. Available input information

Input data was needed for all the binary systems in order to develop the phase diagrams based on the CALPHAD.

For the Bi-Pr system, the phase equilibria data was only measured in the Pr-rich part (up to 7 at% Bi) by Griffin et al. [33] using the methods of X-ray diffraction, thermal, chemical and metallographic analyses. Yoshihara et al. [34] studied the structure data of Pr_5Bi_3 and Pr_2Bi while Gambino et al. [35] reported the crystal structure of Pr_4Bi_3. The solubility data for Pr in liquid Bi at the temperature range of 300–700 °C was determined by Schweitzer et al. [36]. Regarding the results obtained by Refs. [33,34] and assuming the similarity of Bi-Pr to Bi-La and Bi-Nd systems, Moffatt et al. [37] drew a complete phase diagram. In the whole composition range of the Bi-Pr system, five intermetallic compounds (PrBi, $PrBi_2$, Pr_4Bi_3, Pr_5Bi_3, and Pr_2Bi) and three terminal solution phases (rhombohedral (Bi), DHCP (Pr), and BCC_A2 (Pr)) were reported. There are two eutectic reactions of the L ↔ rhombohedral (Bi) + $PrBi_2$ at 544 K and L ↔ Pr_2Bi + BCC_A2 (Pr) at 1167 K. Also, four peritectic reactions of the L + PrBi ↔ $PrBi_2$ at 1773 K, L + PrBi ↔ Pr_4Bi_3 at 1923 K, L + Pr_4Bi_3 ↔ Pr_5Bi_3 at 1498 K, and L + Pr_5Bi_3 ↔ Pr_2Bi at 1373 K exist. The solubility of Bi in BCC_A2 (Pr) solution phase is about 0.8 at% [33], and this solution phase would be assessed using the sub-regular solution model. However, the other two terminal solution phases were treated as pure

element phases because of their very limited solubility. There are several studies reporting the thermodynamic properties of the Bi-Pr system. Colinet et al. [38] studied the enthalpies of formation of the intermetallic compounds based on the Miedema model [39]. Borsese et al. [40] determined the enthalpies of formation of PrBi and Pr_4Bi_3 at 300 K by the method of direct isoperibol calorimeter. All the data of phase equilibria [33], solubility [36] and enthalpy of formation [38,40] were used as the input information. Due to the extremely limited phase information, digital data from the drawn phase diagram by Moffatt [37] was also used in the optimization.

For the Bi-Ce system, Vogel et al. [41] firstly measured the phase-equilibrium of the Bi-Ce binary system experimentally in which four stable intermetallic compounds were reported: Ce_3Bi, Ce_4Bi_3, CeBi, and $CeBi_2$. Pleasance et al. [42] studied the solubility of Ce in liquid Bi. Yoshihara et al. [34] investigated the whole range of the Bi-Ce alloys using X-ray diffraction and metallographic methods and confirmed the first compound existing at the Ce-rich side is Ce_2Bi but not Ce_3Bi as Vogel et al. [41] reported. Gschneidner et al. [43] assessed the phase diagram for the Bi-Ce binary system based on the previous studies [34,41,42]. In the Bi-Ce binary system, CeBi, $CeBi_2$, Ce_4Bi_3, Ce_5Bi_3, and Ce_2Bi were shown to be the stable compounds. Except for Ce_4Bi_3, which melts congruently at 2013 K, all other four stable compounds are formed through peritectic reactions: L + CeBi ↔ $CeBi_2$ at 1183 K, L + Ce_4Bi_3 ↔ CeBi at 1903 K, L + Ce_4Bi_3 ↔ Ce_5Bi_3 at 1670 K, and L + Ce_5Bi_3 ↔ Ce_2Bi at 1518 K. There are also two eutectic reactions: L ↔ rhombohedral (Bi) + $CeBi_2$ and L ↔ Ce_2Bi + BCC_A2 (Ce) reported at the temperature of 544 K and 1023 K, respectively. The Bi-Ce system consists of three terminal solution phases: rhombohedral (Bi), FCC_A1 (Ce), and BCC_A2 (Ce). The solution phase BCC_A2 (Ce) would be assessed with sub-regular solution model in the present work because of the 1 at% solubility of Bi in Ce [43]. Whereas rhombohedral (Bi), FCC_A1 (Ce) would be treated as pure element phases as the solubility of Bi inside is very limited. In addition, the enthalpies of formation of the intermetallic compounds that were determined by Borzone et al. [44]. In the present optimization, all the experimental data [34,41,42,44] discussed above was used as the input information. Additionally, due to the lack of the phase information beyond 80 at% of Ce from experiments, some digital data was picked up from the assessment by Gschneidner et al. [43] for a better optimization.

For the Bi-Er system, Abulkhaev has drawn a phase diagram using results of DTA, X-ray phase analysis, and microstructural analysis [45], which was reproduced by H. Okamoto [46]. In the Bi-Er binary system, two stable intermetallic compounds: ErBi and Er_5Bi_3 exist. There are two eutectic reactions: L ↔ rhombohedral (Bi) + ErBi at 542 K and L ↔ Er_5Bi_3 + HCP_A3 (Er) at 1434 K and one peritectic reaction: L + ErBi ↔ Er_5Bi_3 at 1700 K. ErBi congruently melts at the temperature of 2080 K. Two terminal solution phases: rhombohedral (Bi) and HCP_A3 (Er) exist in this binary system and HCP_A3 (Er) would be assessed using the sub-regular solution model in the present study. The enthalpy of formation of ErBi was measured by Borzone et al. [47] using direct calorimetry method. Besides, enthalpies of formation of ErBi and Er_5Bi_3 were also determined by EMF method [48,49]. All the data collected was used for the optimization.

4. Results and discussion

Based on the available input data, the evaluation of the thermodynamic parameters was carried out by using the PARROT module in the Thermo-Calc software [50]. The optimization process in the Thermo-Calc software was realized by minimizing the sum of squares of the differences between the experimental values and computed ones. Each piece of the input information was given a certain weight based on the uncertainty of the input data. The interaction parameters were developed through carefully assessing the phase relations in the binary systems. The step by step optimization was employed until the output information including phase diagram agreed well with the input ones. The final thermodynamic parameters and the Gibbs free energy expressions for different phases can be obtained after the optimization, then the phase diagrams can be calculated accordingly in the POLY module of Thermo-Calc. Thermodynamic properties such as enthalpy of formation, entropy of formation, enthalpy of mixing of liquid, activity coefficient and activity can also be obtained based on the assessed Gibbs free energy expressions of different phases.

4.1. Phase diagram development

The calculated phase diagram of the Bi-Pr binary system by the sub-regular solution model and two sub-lattice model is shown in Fig. 1. The figure shows that the assessed phase diagram is in good agreement with the literature data [33,36,37]. The calculated compositions and temperatures for the invariant reactions and the literature data are listed in Table 1 which also shows a great consistence. The interaction parameters were developed by assessing the phase relations in the Bi-Pr binary system based on the calculated and experimental data of phase diagram and thermodynamic properties. The optimized parameters of thermodynamic models are shown in Table 2.

Fig. 2 shows the comparison between the calculated Bi-Ce phase diagram and the available literature data [34,41–43] and they agree well with our assessed phase diagram. The comparisons of compositions and temperatures in the calculated invariant reactions with the experimental data for the Bi-Ce system are shown in Table 3, which shows the accuracy of our optimization. The thermodynamic parameters obtained are given in Table 4.

For the Bi-Er system, the comparison between our assessed phase diagram and the experimental data in literature [45] is shown in Fig. 3. It shows that the liquidus and invariant reactions boundaries are nearly identical to the experimental data. All the calculated invariant reactions in Bi-Er are given in Table 5. In addition, the experimental invariant reactions are also listed and compared in Table 5 where a good consistence is obtained. The evaluated parameters of thermodynamic models and Gibbs free energy expressions for the Bi-Er system are shown in Table 6.

4.2. Thermodynamic study

All the phase diagrams for the Bi-Pr, Bi-Ce, and Bi-Er systems have been optimized and agree well with the experimental information. Gibbs free energy expressions for all phases can be known based on the method of the CALPHAD after optimization. Then the thermodynamic properties can be calculated accordingly.

4.2.1. Heat capacity, enthalpy of formation and entropy of formation

In the optimization, Gibbs free energy of formation of intermetallic compounds is written by equation (9) according to the equation (8)

$$G_f = a + bT = \Delta H_f - T\Delta S_f \tag{9}$$

where ΔH_f and ΔS_f are enthalpy of formation and entropy of formation, respectively. It can be known from equation (9) that a and b are corresponding to ΔH_f and $-\Delta S_f$, and the obtained ΔH_f and ΔS_f in the study are temperature-independent. Thus, heat capacity is 0 for all these three binary systems.

The optimized results for a and b in equations (8) and (9) give

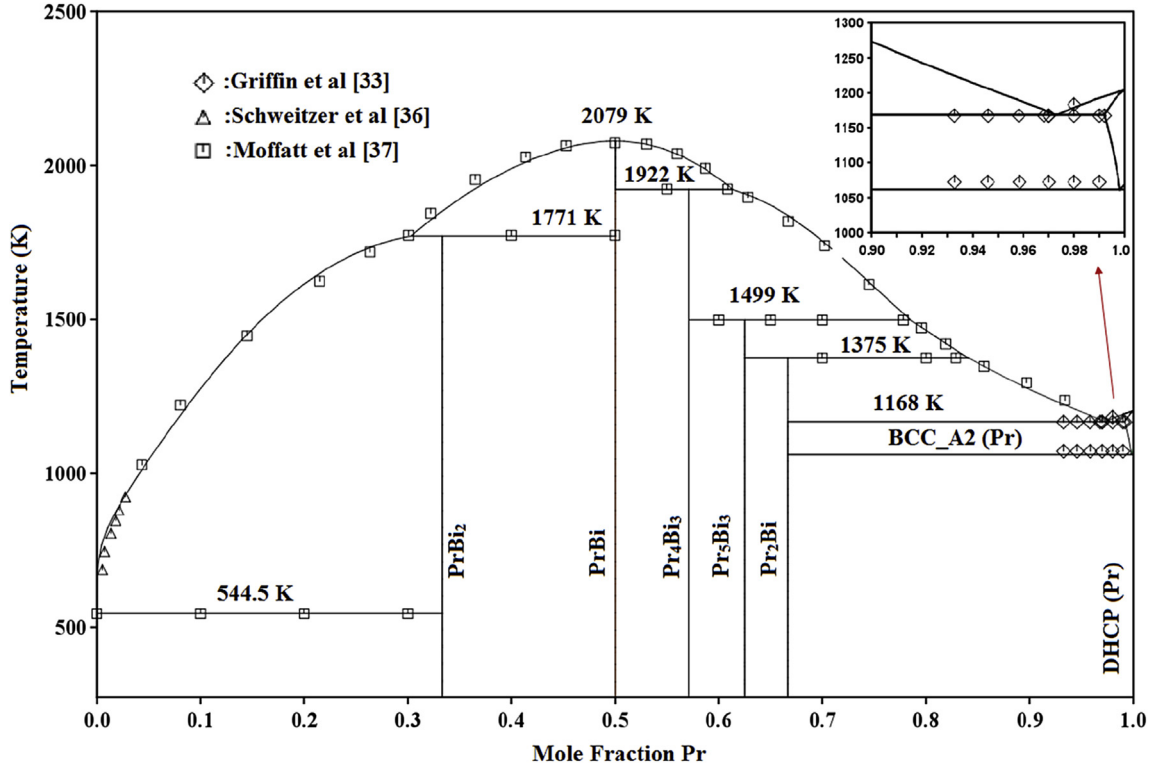

Fig. 1. Calculated Bi-Pr phase diagram compared with literature data [33,36,37].

Table 1
Invariant reactions in the Bi-Pr binary system.

Invariant Reaction	Reaction Type	x^L_{Pr}	T/K	Ref.
L ↔ rhombohedral (Bi) + $PrBi_2$	Eutectic	0.00	544	[37]
		5.2E-6	544.5	This Work
L + PrBi ↔ $PrBi_2$	Peritectic	~0.30	1773	[37]
		0.3031	1771	This Work
L + PrBi ↔ Pr_4Bi_3	Peritectic	~0.61	1923	[37]
		0.6137	1922	This Work
L + Pr_4Bi_3 ↔ Pr_5Bi_3	Peritectic	~0.78	1498	[37]
		0.7851	1499	This Work
L + Pr_5Bi_3 ↔ Pr_2Bi	Peritectic	~0.83	1373	[37]
		0.8410	1375	This Work
L ↔ Pr_2Bi + BCC_A2 (Pr)	Eutectic	~0.97	1167	[33]
		0.9730	1168	This Work

the values of enthalpy of formation and entropy of formation. Enthalpies of formation of the intermetallic compounds in the Bi-Pr, Bi-Ce, and Bi-Er systems together with the results reported in literature are listed in Tables 7–9 respectively. The exothermic enthalpies indicate the heat release when forming these compounds. It can be known from the quantitative comparison with the literature data that the agreement is satisfactory on the whole. Based on the comparison of these three binary systems, a fairly monotonic decrease was found on the value of the Gibbs energy of formation with the increasing atomic number of the lanthanide. The similar phenomenon is also found by other investigators [38]. It is also found that the value of Gibbs energy of formation of the intermetallic compound at equiatomic composition is the largest in one binary system. It means the biggest change of Gibbs energy accompanying the formation of a substance happens at equiatomic composition.

Table 2
Thermodynamic parameters of the Bi-Pr system obtained by the CALPHAD method.

Phase	Thermodynamic Parameters (J/mol)
Liquid (Bi, Pr)	$L_0^{liq} = -7956.29 - 789.84T + 83.32Tln(T)$
	$L_1^{liq} = -2015.64 + 166.08T$
	$L_2^{liq} = 1061.90 - 69.63T$
BCC, formula (Bi, Pr)	$L_0^{BCC} = -37.37 + 151.00T$
	$L_1^{BCC} = -152.31 + 337.21T$
$PrBi_2$	$G^{PrBi_2} = -78697.64 - 5.92T + 0.3333G_{Pr}^{SER,0} + 0.6667G_{Bi}^{SER,0}$
PrBi	$G^{PrBi} = -97648.70 - 7.56T + 0.5000G_{Pr}^{SER,0} + 0.5000G_{Bi}^{SER,0}$
Pr_4Bi_3	$G^{Pr_4Bi_3} = -92876.62 - 8.33T + 0.5714G_{Pr}^{SER,0} + 0.4286G_{Bi}^{SER,0}$
Pr_5Bi_3	$G^{Pr_5Bi_3} = -88136.16 - 4.64T + 0.6250G_{Pr}^{SER,0} + 0.3750G_{Bi}^{SER,0}$
Pr_2Bi	$G^{Pr_2Bi} = -81399.78 - 2.82T + 0.6667G_{Pr}^{SER,0} + 0.3333G_{Bi}^{SER,0}$

The entropies of formation of the intermetallic compounds in the Bi-Pr, Bi-Ce, and Bi-Er systems are given in Table 10. There is no

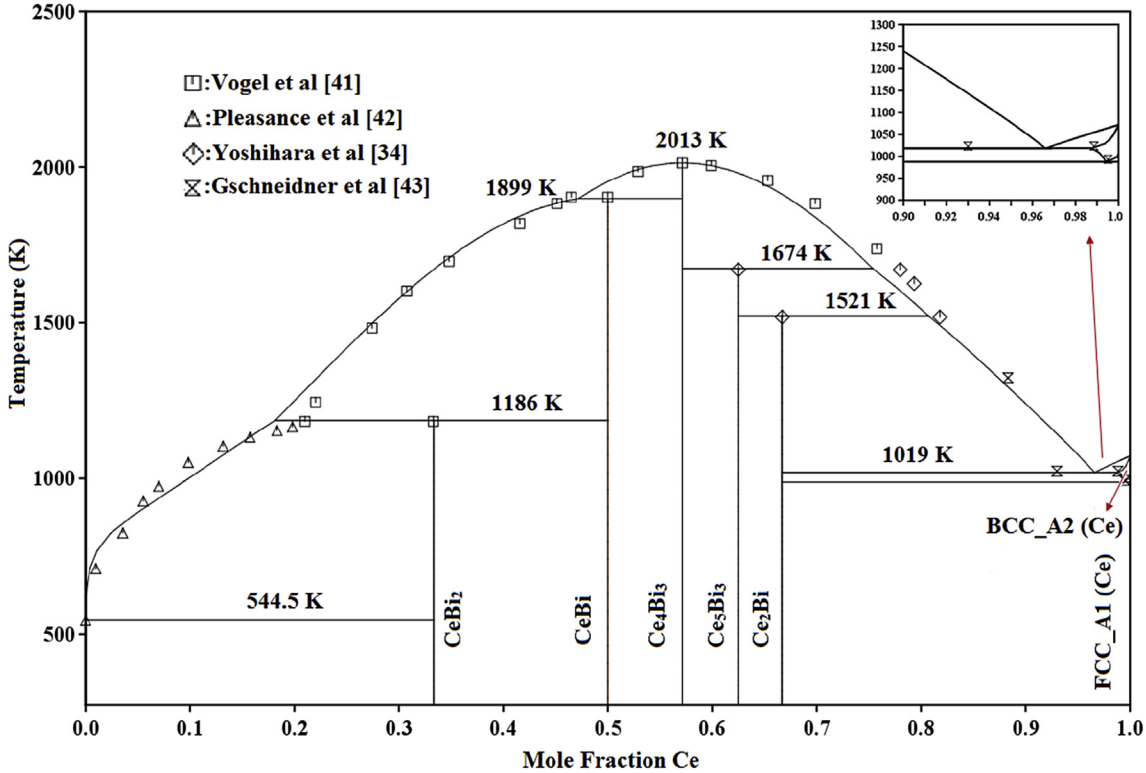

Fig. 2. Calculated Bi-Ce phase diagram compared with literature data [34,41–43].

Table 3
Invariant reactions in the Bi-Ce binary system.

Invariant Reaction	Reaction Type	x^L_{Ce}	T/K	Ref.
L ↔ rhombohedral (Bi) + $CeBi_2$	Eutectic	~0.000	544	[42]
		7.5E-5	544.5	This Work
L + CeBi ↔ $CeBi_2$	Peritectic	~0.21	1183	[41]
		0.1815	1186	This Work
L + Ce_4Bi_3 ↔ CeBi	Peritectic	~0.465	1903	[41]
		0.4721	1899	This Work
L + Ce_4Bi_3 ↔ Ce_5Bi_3	Peritectic	~0.78	1670	[34]
		0.7537	1674	This Work
L + Ce_5Bi_3 ↔ Ce_2Bi	Peritectic	~0.818	1518	[34]
		0.8072	1521	This Work
L ↔ Ce_2Bi + BCC_A2 (Ce)	Eutectic	~0.93	1023	[43]
		0.9659	1019	This Work

available literature data for comparison. However, when considering the values reported for other lanthanides-bismuth binary systems [20–28], our optimization gave realistic and reasonable values for this property.

4.2.2. Enthalpy of mixing of liquid

Fig. 4, Fig. 5, and Fig. 6 show the enthalpies of mixing of the liquid phase for the Bi-Pr, Bi-Ce, and Bi-Er systems respectively at 2500 K in comparison with other literature. The presented thermodynamic data agrees well with the values reported previously. It also can be seen enthalpy of mixing changes with concentration and which decreases firstly and then increases with the increment of concentration. The calculation of enthalpies of mixing at different concentrations greatly extends the experimental data and

Table 4
Thermodynamic parameters of the Bi-Ce system obtained by the CALPHAD method.

Phase	Thermodynamic Parameters (J/mol)
Liquid (Bi, Ce)	$L_0^{liq} = -237286.39 + 6.19T$
	$L_1^{liq} = 3681.43$
	$L_2^{liq} = 46794.67$
BCC, formula (Bi, Ce)	$L_0^{BCC} = -2674.67$
	$L_1^{BCC} = 615475.61$
	$L_2^{BCC} = 441332.95$
$CeBi_2$	$G^{CeBi_2} = -77578.29 + 6.20T + 0.3333G_{Ce}^{SER,0} + 0.6667G_{Bi}^{SER,0}$
CeBi	$G^{CeBi} = -114315.74 + 15.50T + 0.5000G_{Ce}^{SER,0} + 0.5000G_{Bi}^{SER,0}$
Ce_4Bi_3	$G^{Ce_4Bi_3} = -101540.52 + 8.88T + 0.5714G_{Ce}^{SER,0} + 0.4286G_{Bi}^{SER,0}$
Ce_5Bi_3	$G^{Ce_5Bi_3} = -91762.01 + 8.17T + 0.6250G_{Ce}^{SER,0} + 0.3750G_{Bi}^{SER,0}$
Ce_2Bi	$G^{Ce_2Bi} = -81420.73 + 6.36T + 0.6667G_{Ce}^{SER,0} + 0.3333G_{Bi}^{SER,0}$

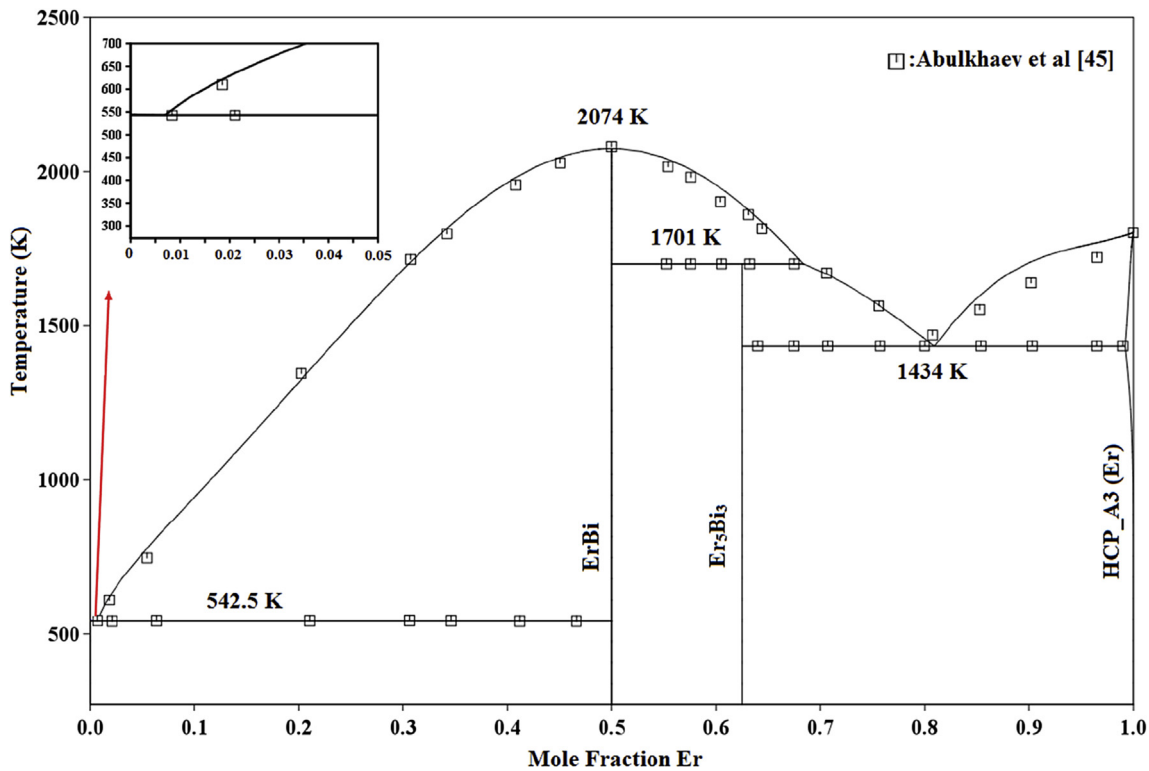

Fig. 3. Calculated Bi-Er phase diagram compared with literature data [45].

Table 5
Invariant reactions in the Bi-Er binary system.

Invariant Reaction	Reaction Type	x_{Er}^{L}	T/K	Ref.
L ↔ rhombohedral (Bi) + ErBi	Eutectic	~8.4E-3	542	[45]
		8.6E-3	542.5	This Work
L + ErBi ↔ Er_5Bi_3	Peritectic	~0.68	1700	[45]
		0.6832	1701	This Work
L ↔ Er_5Bi_3 + HCP_A3 (Er)	Eutectic	~0.80	1434	[45]
		0.8078	1434	This Work

will provide abundant data for the future study.

4.2.3. Activity coefficient and activity

Based on equation (6), the molar excess Gibbs free energy can be written by.

$$\overline{G}_{excess} = x_A x_B \sum_{n=0}^{m} L_n^{liq}(x_A - x_B)^n \tag{10}$$

Table 6
Thermodynamic parameters of the Bi-Er system obtained by the CALPHAD method.

Phase	Thermodynamic Parameters (J/mol)
Liquid (Bi, Er)	$L_0^{liq} = -188097.29$
	$L_1^{liq} = 5429.67$
	$L_2^{liq} = 48809.49$
	$L_3^{liq} = -16792.89$
HCP, formula (Bi, Er)	$L_0^{BCC} = -104311.13$
ErBi	$G^{ErBi} = -80350.02 + 1.93T + 0.5000G_{Er}^{SER,0} + 0.5000G_{Bi}^{SER,0}$
Er_5Bi_3	$G^{Er_5Bi_3} = -66485.80 + 2.54T + 0.6250G_{Er}^{SER,0} + 0.3750G_{Bi}^{SER,0}$

Table 7
Enthalpies of formation of the intermetallic compounds in the Bi-Pr system.

Compositon	x_{Pr}	ΔH_f (kJ/mol)		
		Colinet et al. [38]	Borsese et al. [40]	This Work
$PrBi_2$	0.333	−79	–	−78.698
PrBi	0.500	−97	−100	−97.649
Pr_4Bi_3	0.571	−94	−94	−92.877
Pr_5Bi_3	0.625	−88	–	−88.136
Pr_2Bi	0.667	−81	–	−81.400

Table 8
Enthalpies of formation of the intermetallic compounds in the Bi-Ce system.

Compositon	x_{Ce}	ΔH_f (kJ/mol)		
		Colinet et al. [38]	Borzone et al. [44]	This Work
$CeBi_2$	0.333	−84	−77.5	−77.578
CeBi	0.500	−102	−114.7	−114.32
Ce_4Bi_3	0.571	−99	−101.3	−101.54
Ce_5Bi_3	0.625	−92	−91.7	−91.762
Ce_2Bi	0.667	−84	−81.6	−81.421

Table 9
Enthalpies of formation of the intermetallic compounds in the Bi-Er system.

Compositon	x_{Er}	ΔH_f (kJ/mol)				
		Colinet et al. [38]	Petrashgkevich et al. [48]	Borzone et al. [47]	Yamshchikov et al. [49]	This Work
ErBi	0.500	−75	−109	−90	−103	−80.350
Er_5Bi_3	0.625	−69	≤ −81	–	≤ −77	−66.486

Table 10
Entropies of formation of intermetallic compounds for the Bi-Pr, Bi-Ce, and Bi-Er systems.

Compositon	x (fission products)	ΔS_f (J/mol)
$PrBi_2$	0.333	5.92
PrBi	0.500	7.56
Pr_4Bi_3	0.571	8.33
Pr_5Bi_3	0.625	4.64
Pr_2Bi	0.667	2.82
$CeBi_2$	0.333	−6.20
CeBi	0.500	−15.50
Ce_4Bi_3	0.571	−8.88
Ce_5Bi_3	0.625	−8.17
Ce_2Bi	0.667	−6.36
ErBi	0.500	−1.93
Er_5Bi_3	0.625	−2.54

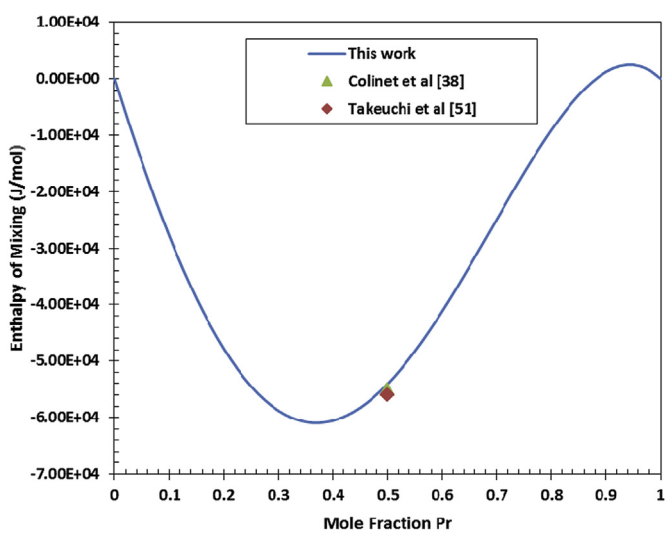

Fig. 4. The calculated enthalpy of mixing of the liquid phase in the Bi-Pr system compared with literature data [38,51].

Then the activity coefficient of A, for example, can be calculated by.

$$RTln\gamma_A = \left(\frac{\partial(n_A + n_B)\overline{G}_{excess}}{\partial n_A}\right)_{n_B,p,T} \quad (11)$$

where n is mole number. The activity coefficients of Pr in liquid bismuth at different concentrations under the temperatures of 873, 923 and 973 K are shown in Fig. 7. For all the calculations of the activity coefficients in this study, the reference states were supercooled states. Fig. 7 shows that the activity coefficient of Pr in liquid bismuth increases with the increment of concentration and temperature. The value of the activity coefficient of Pr in liquid bismuth is extremely small and in the order of magnitude of 10^{-13} to 10^{-11} at the temperature range of 873 K–973 K. It is a little smaller than the value with the order of magnitude of 10^{-12} at the temperature of 823 K reported by Zhang et al. [53] previously. However, if

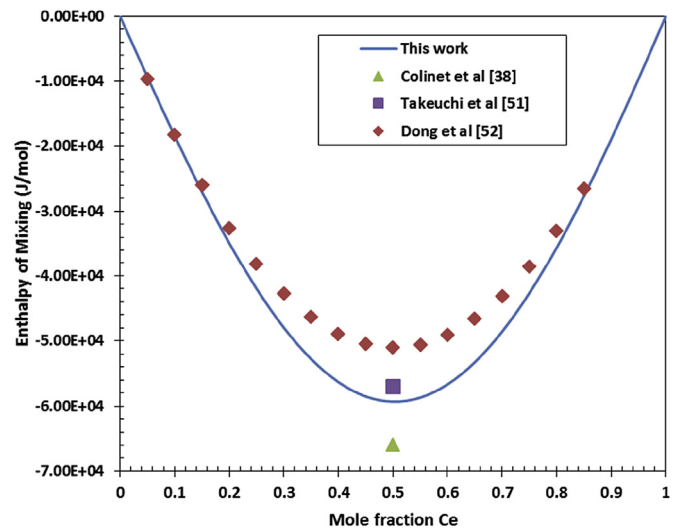

Fig. 5. The calculated enthalpy of mixing of the liquid phase in the Bi-Ce system compared with literature data [38,51,52].

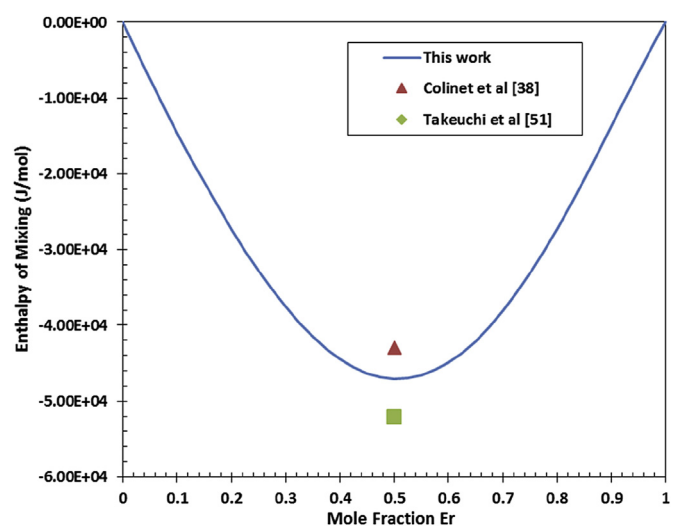

Fig. 6. The calculated enthalpy of mixing of the liquid phase in the Bi-Er system compared with literature data [38,51].

considering the value of activity coefficient is extremely small at this concentration and temperature ranges, this discrepancy is acceptable.

Fig. 8 presents the activity coefficient of Ce in liquid bismuth at different concentrations under the temperatures of 873, 923 and 973 K. It shows a slight decrease with the increase of mole fraction of Ce in liquid bismuth at low concentration. The value of activity coefficients starts to increase gradually at the mole fraction around 0.02 at%. In addition, temperature also plays an important role on the activity coefficient, and the value of which at higher

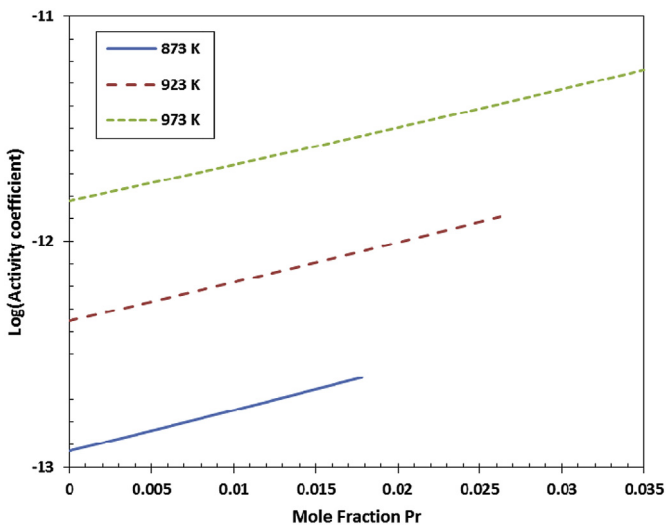

Fig. 7. Activity coefficient of Pr in liquid Bi at different concentrations under three temperatures: 873, 923, and 973 K.

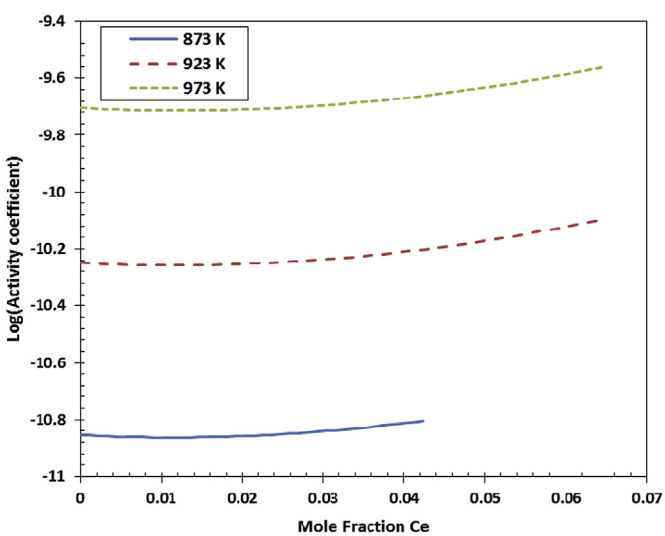

Fig. 8. Activity coefficients of Ce in liquid Bi at different concentrations under three temperatures: 873, 923, and 973 K.

Table 11
Comparison between the calculated activity coefficients and literature values for the Bi-Ce system.

Temperature	Concentration	Log(activity coefficient)	Ref.
873 K	Dilution	−11.03	[54]
	Dilution	−11.31	[53]
	0–4.23 at%	−10.85~-10.80	This Work
923 K	Dilution	−10.32	[54]
	Dilution	−10.61	[53]
	0–6.42 at%	−10.25~-10.10	This Work

temperature is larger than the one at lower temperature for the same concentration. The calculated activity coefficients at the temperatures of 823 K and 873 K are compared with the literature values reported before and shown in Table 11, from which it can be seen the calculated values are close to the literature data. It verified the reliability of our calculated activity coefficients at different concentrations and temperatures.

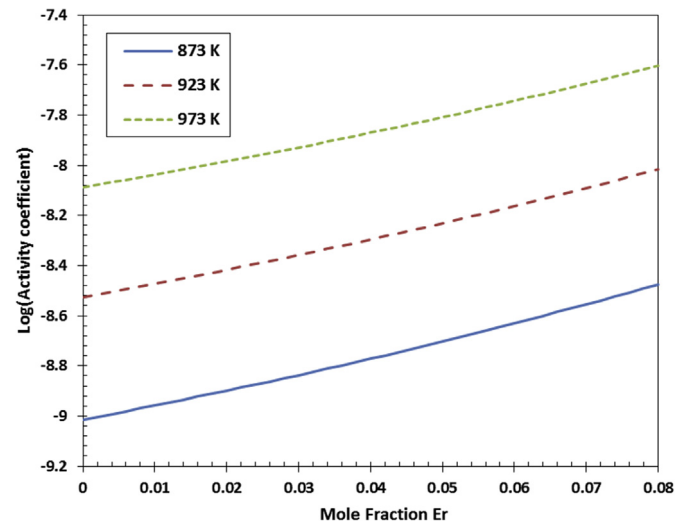

Fig. 9. Activity Coefficients of Er in liquid Bi at different concentrations under three temperatures: 873, 923, and 973 K.

The activity coefficients of Er in liquid bismuth at different concentrations under the temperatures of 873, 923 and 973 K were also investigated in the present work. Fig. 9 presents our calculated results of the activity coefficients of Er in liquid bismuth at the concentration range of 0–8 at%. It can be seen the activity coefficient increases with the increment of mole fraction of Er in liquid bismuth and temperature as well. The calculated activity coefficients are compared with the literature data in Table 12, which shows an excellent agreement.

Figs. 7–9 show that the dependence of the activity coefficient of the element on its concentration is different for different element. Based on equations (10) and (11), it can be found that interaction parameter L_i^{liq} and maximum power m affect the concentration-dependence of the activity coefficient. L_i^{liq} works as a constant and is independent on the mole fraction of species. However, the maximum power m sufficiently to fit a system may be different for different cases in the practical optimization. It explains why the dependence of the activity coefficients on the concentrations for Bi-Pr, Bi-Ce, and Bi-Er is different.

Activity is product of activity coefficient and mole fraction.

$$a = \gamma x \tag{12}$$

Because of the similar patterns of the activity coefficients at three temperatures for each lanthanide, here only the activities at 973 K were plotted and analyzed to avoid the redundancy. The activities of the three elements considered are shown in Fig. 10. It is shown that under these concentration ranges calculated, the activity is far away from the Raoult's law [57,58], which should be a unit-slope diagonal line in the coordinate system. It indicates dramatically negative deviations of these systems from the ideal ones. The interactions between unlike atoms should be therefore significantly different from these between like atoms. Generally more energy should be released for the attraction set up between unlike atoms than that needed to break the original atom pairs, which results in an exothermic enthalpy of mixing. This conclusion is consistent with the negative enthalpy of mixing in Figs. 4–6.

If the low concentration part is focused, the activities show linear relationships up to concentrations of 0.8%, 2.0%, and 1.6%, respectively, for the Ce, Pr, and Er. The phenomenon can be explained by Henry's law [59], which states

Table 12
Comparison between the calculated activity coefficients and literature values for the Bi-Er system.

Temperature	Concentration	Log(activity coefficient)	Ref.
873 K	Dilution	−9.21	[55]
	Dilution	−9.14	[56]
	0–8 at%	−9.01~-8.47	This Work
923 K	Dilution	−8.62	[55]
	Dilution	−8.73	[56]
	0–8 at%	−8.52~-8.02	This Work
973 K	Dilution	−8.08	[55]
	Dilution	−8.37	[56]
	0–8 at%	−8.09~-7.60	This Work

$$a_{Ln} = x_{Ln} K_{Ln} \tag{13}$$

where a_{Ln} is the activity of the lanthanide, K_{Ln} is the Henry's constant. Using pure lanthanide as the reference, K_{Ln} is related to the limiting activity coefficient by

$$\gamma_{Ln}^{\infty} = K_{Ln} \big/ a_{Ln}^{0} \tag{14}$$

where a_{Ln}^{0} is the activity of pure lanthanide, which is unity. It indicates a constant activity coefficient and a linear relationship between activity and concentration. However, beyond these concentrations, neither Raoult's law nor Henry's law is followed. The activity increases faster than a linear relationship and shows a dependence on the concentration. Further studies theoretically or experimentally are needed to understand the properties thoroughly.

4.2.4. Solubility

The solubility of Pr, Ce, and Er in liquid bismuth at the temperature range of 873 K–973 K was also studied in this work. Generally, the solubility S in the unit of mole fraction can be expressed by Ref. [60].

$$logS = k_1 + k_2/T \tag{15}$$

where k_1 and k_2 are constants. The data of solubility was selected from the calculated phase boundary where $PrBi_2$ and liquid coexist in the Bi-Pr system, $CeBi_2$ and liquid coexist in the Bi-Ce system, and ErBi and liquid coexist in the Bi-Er system. Fig. 11 shows the calculated solubility of Pr, Ce, and Er in liquid bismuth together with the fitting curve, then the correlation between $logS$ and $1/T$ is given in Table 13. The correlation obtained in this study can bring much convenience to other investigators to get the solubility at the desired temperature.

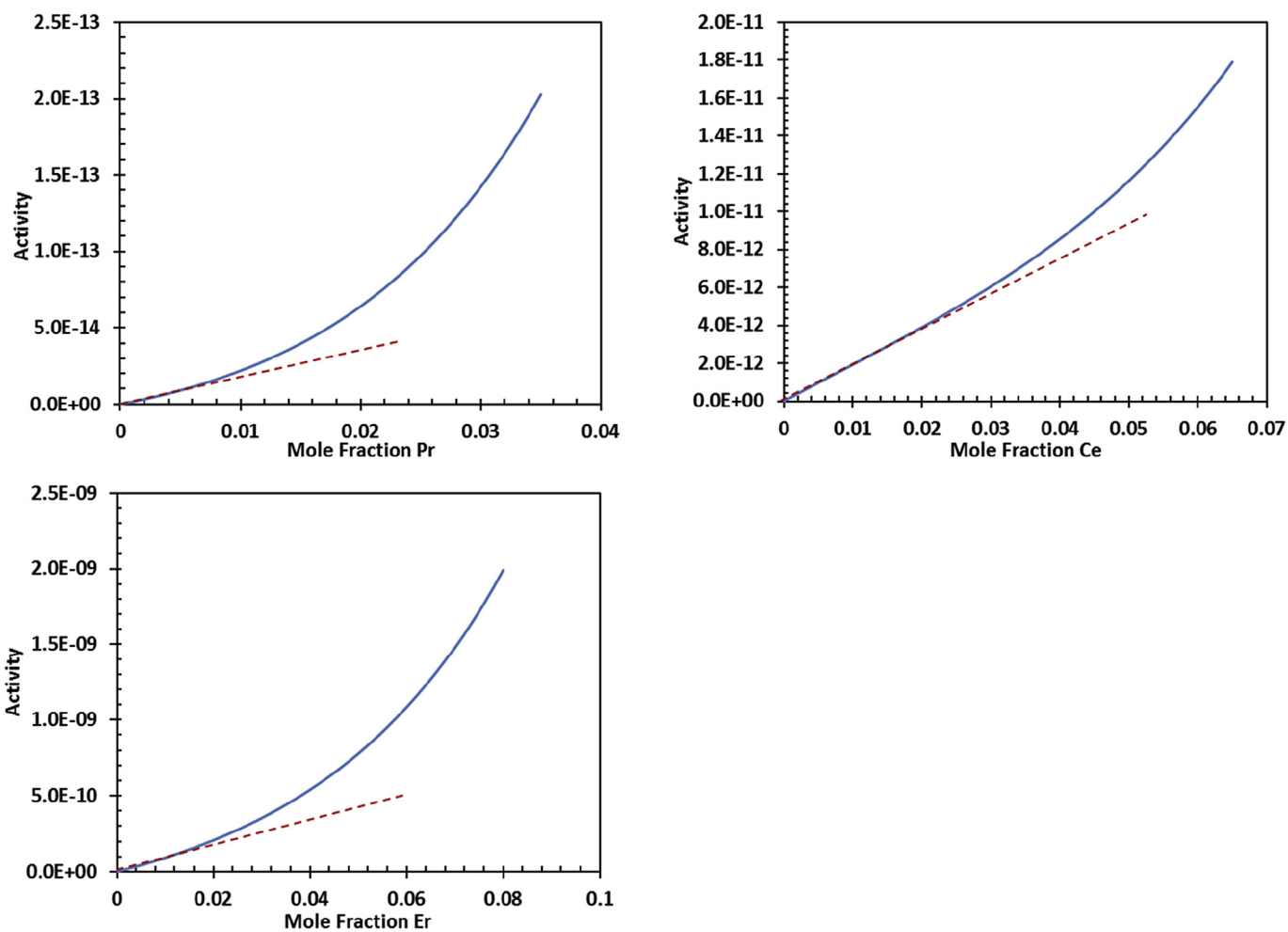

Fig. 10. Activity of Pr, Ce, and Er in liquid bismuth at the temperature of 973 K. Solid line is the activity at different concentrations while the dashed line is the linear relationship of the activity with concentration at dilution.

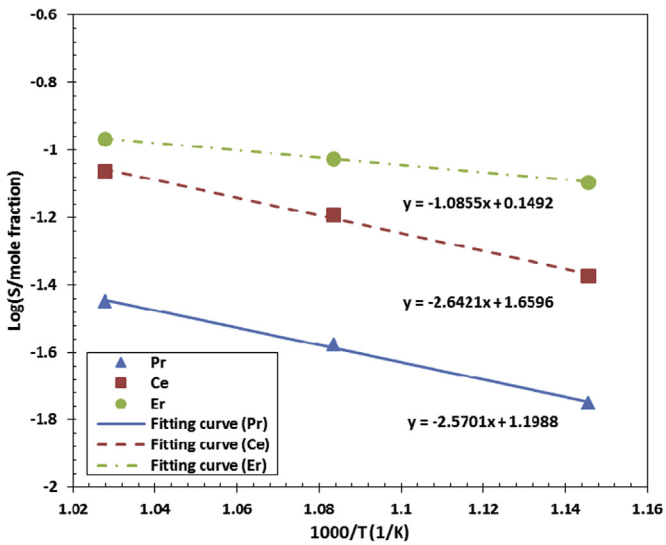

Fig. 11. Calculated solubility of Pr, Ce, and Er in liquid bismuth with the fitting curve.

Table 13
Solubility of Pr, Ce, and Er in liquid bismuth.

Elements	Solid in equilibrium with the liquid	Fitting correlation
Pr	$PrBi_2$	$logS = 1.1988 - 2570.1/T$
Ce	$CeBi_2$	$logS = 1.6596 - 2642.1/T$
Er	ErBi	$logS = 0.1492 - 1085.5/T$

5. Summary and conclusions

Thermodynamic assessments for the Bi-Pr, Bi-Ce, and Bi-Er binary systems have been carried out through the CALPHAD method using sub-regular model and the two-sublattice model. These three binary systems were mainly evaluated based on the available phase information and other thermodynamic properties published in literature. The calculated phase diagrams show a great consistence with the experimental ones. Entropies and enthalpies of formation for metallic compounds and enthalpies of mixing for liquid phase were obtained after the acquirement of thermodynamic parameters in the Gibbs free energy expressions for all phases. The assessed enthalpies of formation and enthalpies of mixing were illustrated to have a good agreement with the literature values. The study about the enthalpies of mixing for liquid phase greatly extends the existed experimental data.

The solubility of fission products (Pr, Ce, and Er) in liquid bismuth and the correlations between solubility and temperature were obtained. Furthermore, the activity coefficients of fission products at different concentrations and temperatures were also evaluated based on the assessed thermodynamic parameters. It is found that the activity coefficient is a function of temperature and concentration, and it increases with the element concentration and temperature on the whole. The change pattern of the activity with concentration were also analyzed and discussed which greatly enhanced our understanding of the three binary systems.

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